

FINAL YEAR PROJECT · TEAM REPORT

OrbitAvanya Email Automation System

An end-to-end cold-email outreach engine for **OrbitAvanya Tech LLP** — sends validated, personalized emails; tracks opens, clicks, visits and time-on-site; detects replies and bounces over IMAP; sends multilingual follow-ups; and reports everything into a color-coded Excel master tracker.

~60 FEATURES BUILT	30+ LANGUAGES	4 PYTHON SCRIPTS	\$0 PAID APIS	24×7 TRACKING
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— 01 · PROJECT OVERVIEW

Why this exists

Manually emailing hundreds of CEOs doesn't scale — and gives zero feedback. OrbitAvanya automates the full outreach lifecycle end-to-end: **validate** → **send** → **track** → **follow-up** → **report**.

Problem

- **Slow** — copy/paste at scale is impossible
- **No feedback** — no idea who opened or clicked
- **High risk of mistakes** — duplicates, fakes, missed follow-ups
- **No visibility** — no bounce detection, no location insight

Solution

- Reads a bulk lead spreadsheet, **validates every email** (4 free layers)

- Sends **personalized HTML** with open / click / visit / time-on-site tracking
- Sends in the lead's **detected or preferred language**
- Watches the inbox, flags **replies & bounces**, auto-reminds non-responders
- Runs **hands-free in batches** — pauses, checks, resumes automatically

IN SCOPE

- Automated bulk SMTP sending
- Multi-layer email validation — zero cost
- Open / click / visit / time-on-site tracking
- Reply & bounce detection over IMAP
- Automatic follow-up reminders
- Duplicate protection + AUTO-CONTINUE batching
- Localized / multilingual emails (30+ languages)
- Excel-based reporting (tracker + report file)

OUT OF SCOPE

- Paid email-deliverability services
- Full inbound CRM
- Legal / compliance advisory — leads are pre-scraped; sending at scale must still respect anti-spam law in the target region

What Existed Before

- A Jupyter notebook that could send basic emails via SMTP
- A single 1×1 open-tracking pixel (server-side only)
- Manual per-run operations, no batching or validation

EXISTING

What We Built (NEW)

- One-file all-in-one launcher `RUN_CAMPAIGN.py`
- 4-layer email validation & fake-email quarantine sheet
- Full click / visit / time-on-site funnel + PHP backend
- Multilingual engine + 2-day auto follow-up

NEW · ~60 FEATURES

What We Can Release Next

- Interactive web dashboard & charts
- Multi-mailbox / multi-campaign management
- Unsubscribe (suppression) management
- Cloud scheduler — track without laptop on

ROADMAP

— 02 · EXISTING SYSTEM

What already existed before this project

The final-year project did **not** start from a blank page. A first iteration existed as an exploratory notebook; this project hardened it into a production-ready system.

BEFORE — V0 NOTEBOOK PROTOTYPE

What was already there

- Jupyter notebook printing **basic SMTP sends** (login → send → log)
- A **single open-tracking pixel** hitting a local Flask server (port 5050)
- A simple **duplicate guard** JSON file to avoid double-sends
- Basic reply scanning and a follow-up subject placeholder
- Lead data already scraped into **Excel (22 columns)**

All code, one notebook, manual restarts, no validation, no reporting.

AFTER — V1 PRODUCTION SYSTEM

What we added

- **One-file launcher** — installs libs, starts server, sends, tracks
- **4-layer validation** with an auto Fake CEO Email quarantine sheet
- **Click, visit, time-on-site** tracking + geolocation + Google-proxy filter
- **Production PHP tracker** on cPanel — works even when laptop is off
- **Multilingual** emails (30+ languages) with brand-name protection
- **Auto follow-ups** (2-day reminders) to non-responders
- **Reply/bounce detection** over IMAP, every 5 minutes
- **6-sheet Excel master tracker** + auto-refreshed report workbook

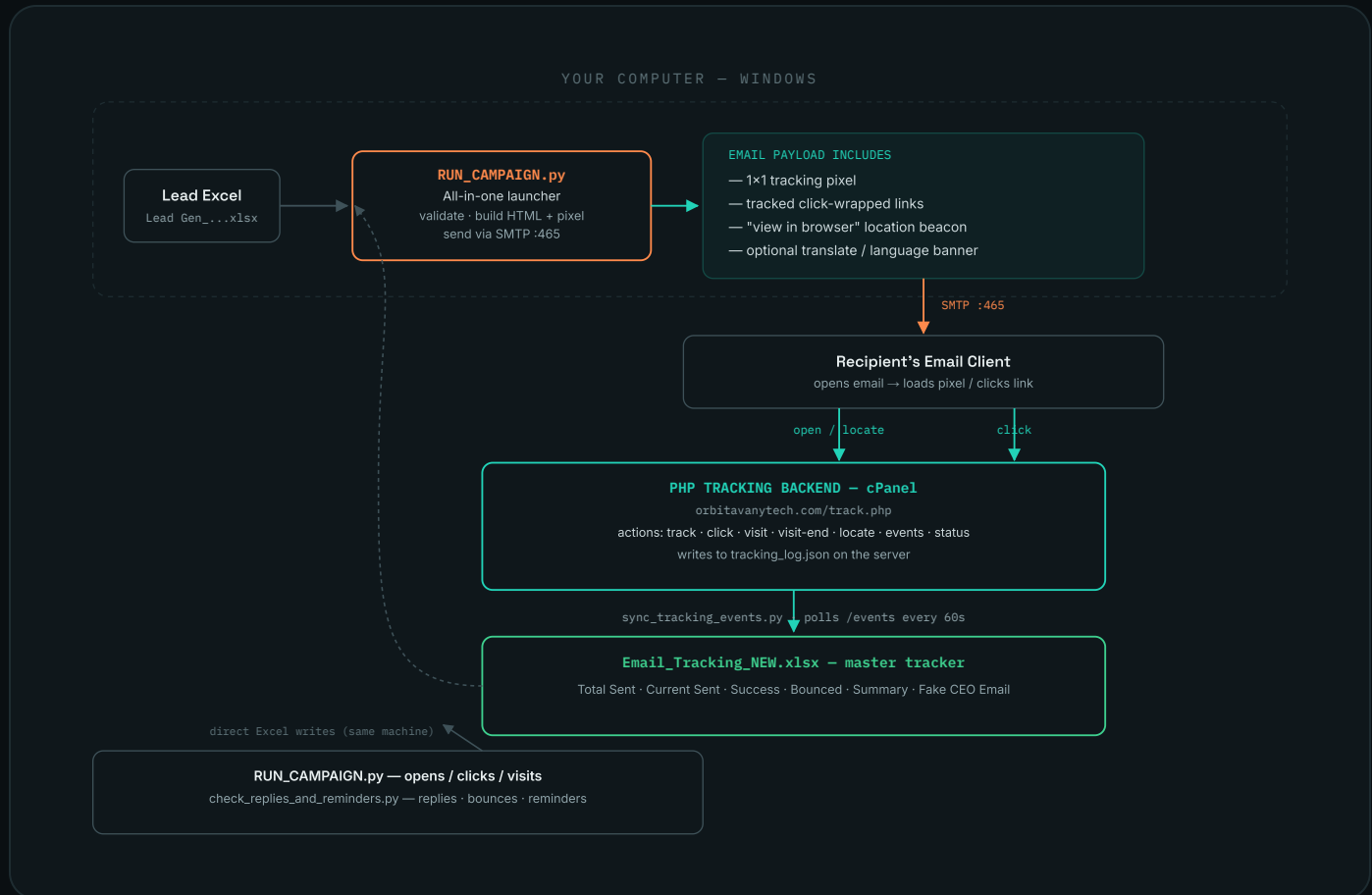
Component-wise: Existing vs New

Component	Existing (Before)	New (This Project)
Sending	Notebook, manual SMTP login per run	Reused SMTP session, auto-install, one-command launcher
Validation	None — emails sent blindly	NEW Syntax → disposable → MX → SMTP mailbox probe
Tracking	1 pixel → Flask → Excel	NEW Pixel + tracked links + visit page + time-on-site + geolocation + PHP prod endpoint
Language	Static English only	NEW 30+ languages, India regional states/cities, brand-name protection
Follow-ups	Manual / placeholder	NEW Auto reminder 2 days after send, location-aware language
Replies & bounces	Manual inbox checks	NEW IMAP scanning every 5 min, auto detect & copy to Bounced sheet
Reporting	Raw logs	NEW 6-sheet tracker + Summary + Sent/Bounced report + date-wise totals
Reliability	Requires laptop + manual restart	NEW AUTO-CONTINUE batches, dedupe JSON, thread-lock file safety, retry-on-lock

03 · SYSTEM ARCHITECTURE

How the pieces fit together

One launcher script drives everything — reading leads, validating, sending, and standing up a local tracker — mirrored in production by a PHP endpoint that survives your laptop going offline.



PATH A — LOCAL FLASK (DEV/FALLBACK)

- Embedded Flask app on port 6060 — routes /track /click /visit /visit-end /locate /status /events
- Exposed publicly with **cloudflared** quick tunnel (*.trycloudflare.com)
- Named-tunnel option for a fixed domain
- Updates Excel **directly** with a thread lock

PATH B — PHP ON CPANEL (PRODUCTION)

- `Server/track.php` on `public_html/track.php` — always-on hosting
- Recipient actions hit `track.php?action=...` and are written to `tracking_log.json`
- Geolocation via `ip-api.com` (free)
- `sync_tracking_events.py` pulls & applies events to Excel every 60s

Why both? The PHP endpoint survives your laptop being offline — recipients can open emails at any hour. The Flask path gives instant local writes during development. When `TRACKING_SCRIPT_URL` is set, the PHP path + sync script becomes the production mechanism.

Module Responsibilities

MODULE	RESPONSIBILITY
<code>RUN_CAMPAIN.py</code>	Everything — config, validation, sending, embedded Flask tracker, Excel writes, reply/bounce, reminders, summaries
<code>email_tracker_server.py</code>	Standalone open-tracking server (port 5050) — original design, kept for reference/dev
<code>check_replies_and_reminders.py</code>	Runs reply/bounce + reminder logic continuously in its own terminal
<code>sync_tracking_events.py</code>	Pulls PHP-side events into Excel every 60 seconds
<code>Server/*.php</code>	Public tracking backend — pixel, click, visit, time-on-site, location, events API
<code>Templates/*.html</code>	Email bodies with <code>{{CEO_Name}}</code> / <code>{{Company_Name}}</code> + portfolio JS snippet
<code>Data/</code>	Lead lists, master tracker, JSON logs, reports
<code>Tools/</code>	<code>cloudflared.exe</code> , <code>ngrok.exe</code>

Data Flow — From Lead to Report

1

Read

`pd.read_excel(Config.EXCEL_FILE_PATH)` loads the lead sheet

2

Enrich (auto)

Missing Country/City resolved from the company email domain → IP → geolocation; sender location is the final fallback

3

Validate

4-layer email check. Fakes go to the Fake CEO Email sheet and are skipped

4

Send

Personalization, optional language banner/full translation, pixel + click-wrap injection, SMTP over a shared connection

5

Track

Recipient actions hit the PHP endpoint → events → `sync_tracking_events.py` → Excel (or directly via Flask)

6

Watch

IMAP scans every 5 min → replies & bounces flagged in the tracker

7

Remind

After 2 days, non-responders automatically receive the follow-up template

8

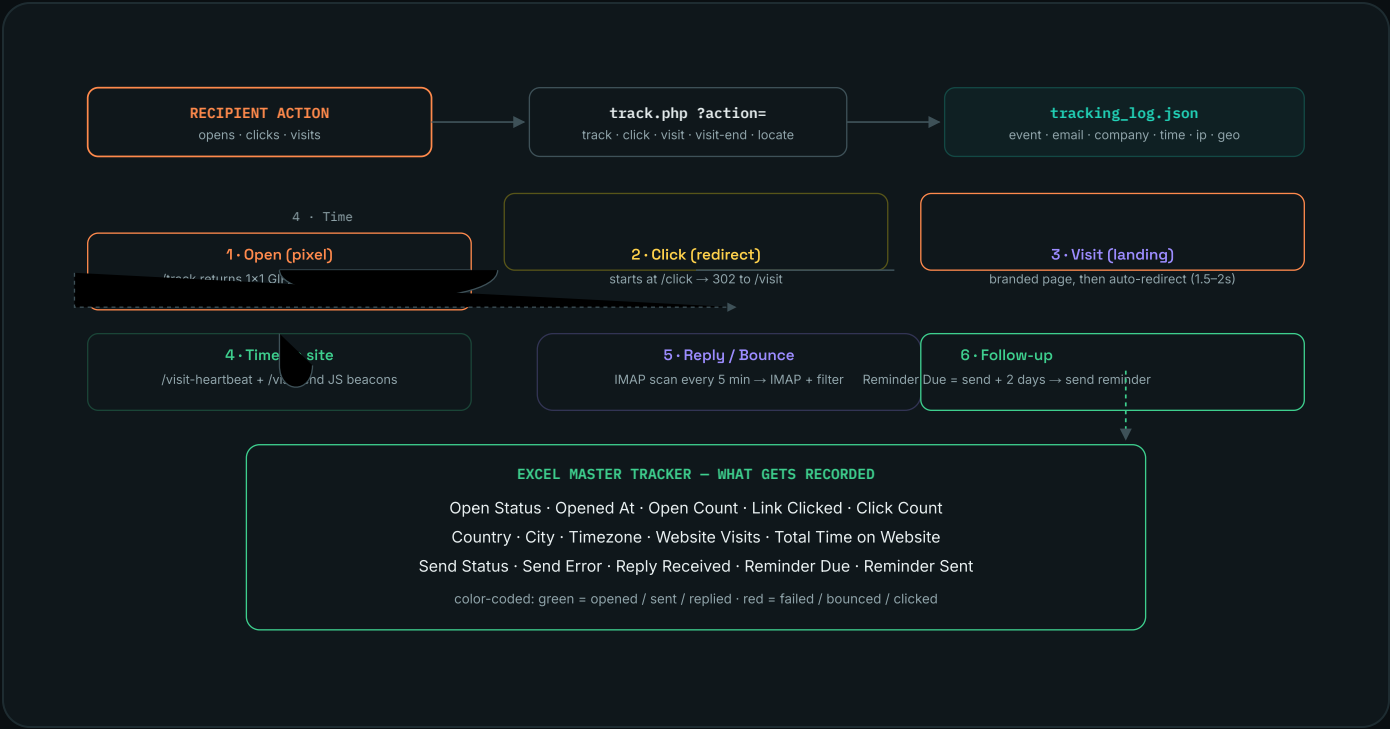
Report

`Sent_Bounced_Report.xlsx` rebuilt after every send / reply / bounce / reminder action

04 · TRACKING & FUNNEL DESIGN

How sends become data

Every email carries a pixel, a click-redirect and a beacon; the funnel below turns a click into a rich Excel signal in seconds.



Data safety: all Excel writes go through `threading.Lock`, records are flushed in batches of 10, duplicate protection is JSON-backed, and the master *Total Sent* sheet is append-only — nothing is ever overwritten.

05 · FEATURES MADE NEW

Everything the system now does

~60 features delivered across eight categories. All marked NEW — every one of them built in this project.

A

Email Sending

#	FEATURE	WHERE
1	Personalized HTML emails — fills <code>{{CEO_Name}}</code> / <code>{{Company_Name}}</code>	NEW
2	Reused SMTP connection (auto-reconnect + retry on drop)	NEW
3	Fully-formed RFC headers — Message-ID, Date, List-Unsubscribe (+ One-Click)	NEW
4	Plain-text + HTML multipart for all clients	NEW
5	Random delay between sends (reputation-safe)	NEW
6	Batch limit + AUTO-CONTINUE pausing / resuming	NEW
7	Duplicate protection via <code>sent_emails_log.json</code>	NEW
8	Failure recording with error message in tracker	NEW

B

Email Validation

FREE · NO PAID API

#	FEATURE	WHERE
9	Syntax check — regex	<code>_syntax_ok()</code>
10	Disposable-domain blocklist — 30+ temp mailing hosts	<code>DISPOSABLE_DOMAINS</code>
11	MX record check (dnspython)	<code>_has_mx()</code>

12	SMTP mailbox probe — MAIL FROM / RCPT TO, retries greylisting	<code>_smtp_mailbox_check()</code>
13	Unknown verdict still sends (max reach)	<code>validate_email()</code>
14	Fake emails stored in <code>Fake CEO Email</code> sheet with reason	<code>save_fake_email_record()</code>

C

Tracking

pixel · click · visit · time-on-site · location

#	FEATURE	WHERE
15	Invisible 1×1 tracking pixel	<code>inject_pixel()</code>
16	Open status, first-open timestamp, open count	<code>update_open_status()</code>
17	Click-Wrapped orbitavanya.com links → /click	<code>wrap_links_with_tracking()</code>
18	Real location on click — country / city / timezone (Cloudflare-aware IP)	<code>get_location_from_ip()</code>
19	Google-proxy IP filter (crawler noise)	<code>GOOGLE_IP_RANGES</code>
20	Website-visit landing + visit count / timestamps	<code>visit_page()</code>
21	Time-on-website via JS heartbeat + end beacons	<code>visit_heartbeat()</code> / <code>visit_end()</code>
22	Location beacon — "view in browser" link	<code>build_location_beacon()</code>
23	Production PHP endpoint (track.php) on cPanel	<code>Server/track.php</code>
24	cloudflared quick tunnel + named-tunnel option	<code>get_cloudflared()</code>
25	Standalone Flask tracker (port 5050)	<code>email_tracker_server.py</code>

D

Replies & Bounces

#	FEATURE	WHERE
26	IMAP reply detection every 5 min (sender match)	<code>check_replies()</code>

27	Bounce detection (mailer-daemon / delivery-failure)	<code>check_replies()</code>
28	Rolling 30-day inbox window	<code>mail.search(SINCE)</code>
29	No double-counting — replied/bounced leads skipped	<code>check_replies()</code>

E

Follow-up Reminders

#	FEATURE	WHERE
30	Automatic Reminder Due = sent date + 2 days	<code>save_email_record()</code>
31	Only to non-responders (skips replied/bounced/reminded)	<code>send_due_reminders()</code>
32	Location-aware reminder language	<code>send_due_reminders()</code>
33	Runs in both launcher loop and standalone script	both scripts

F

Multilingual & Localization

#	FEATURE	WHERE
34	30+ country → language mapping	<code>COUNTRY_LANGUAGE</code>
35	India regional city + state language maps	<code>INDIA_CITY_LANGUAGE</code>
36	Autos- detect lead location from email domain	<code>detect_client_location_from_domain()</code>
37	HTML body translation (BeautifulSoup) — structure preserved	<code>translate_html_body()</code>
38	Brand / name protection during translation	<code>_protect_terms()</code>
39	"View this email in your language" translate banner	<code>build_translate_banner()</code>
40	Translated /visit landing for visitors	<code>visit_page()</code>
41	Language recorded in Excel column	<code>save_email_record()</code>

G

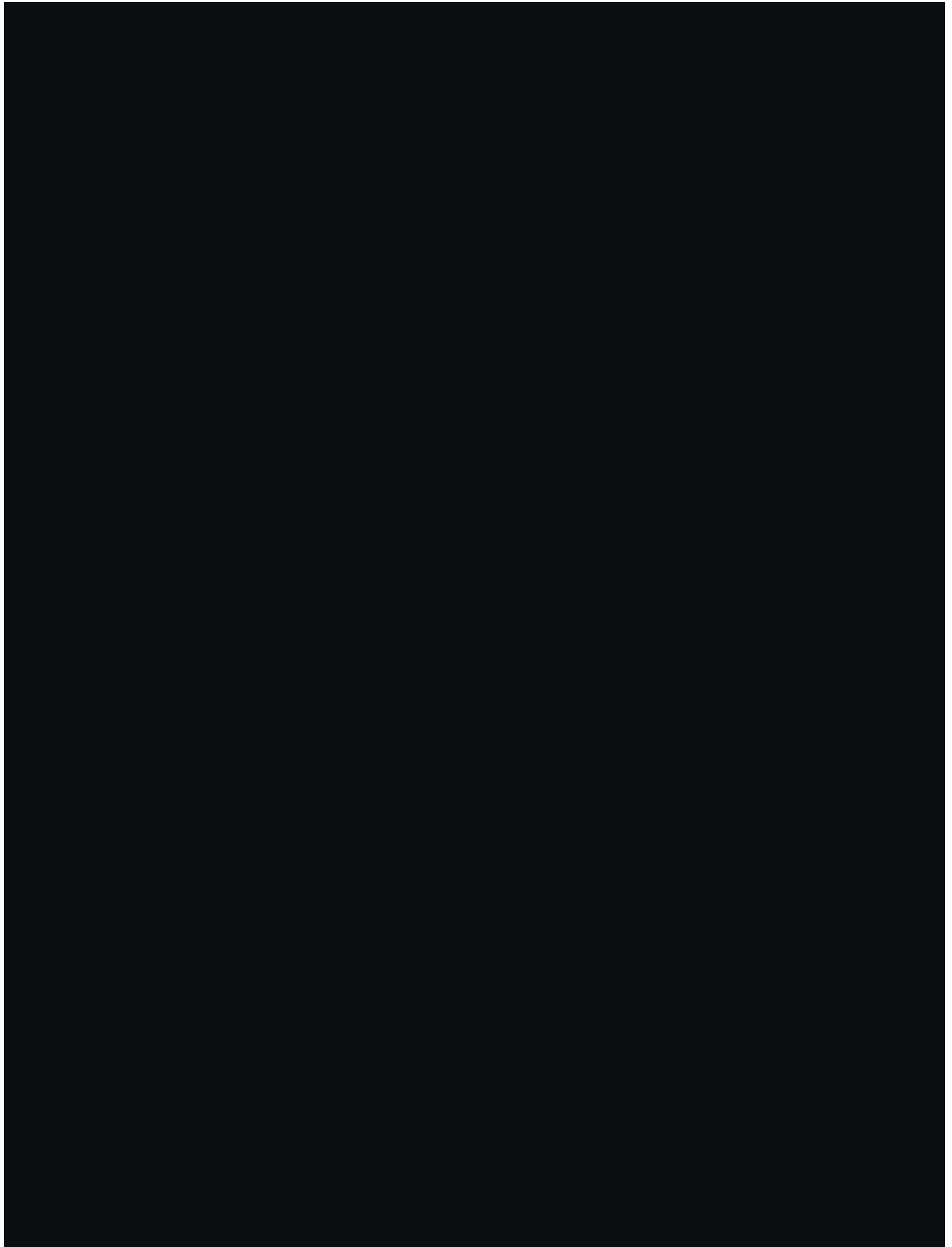
Excel Reporting

#	FEATURE	WHERE
42	Color-coded status cells (green / red)	<code>_style_send_status()</code>
43	6-sheet tracker — Total Sent, Current, Success, ColumnaSummary, Fake	Section 7
44	Append-only master; 15-day prune on Current Sent	<code>flush_saved_records()</code>
45	Update-in-place (no duplicate rows)	<code>flush_saved_records()</code>
46	Cumulative summary + per-date totals	<code>update_campaign_summary()</code>
47	Auto-refreshed Sent/Bounced/Summary report file	<code>regenerate_sent_bounced_report()</code>
48	Batched flushing (fast Excel writes)	<code>PENDING_FLUSH_SIZE</code>

H

Automation & Reliability

#	FEATURE	WHERE
49	Auto install missing libraries on first run	<code>install()</code>
50	Auto-download of cloudflared tunnel client	<code>get_cloudflared()</code>
51	5-step console wizard feedback	entry point
52	tqdm progress bars	<code>run_campaign()</code>
53	Keep-alive loop — replies every 5 min, reminders every 10 min	main block
54	Event JSON log (tracking_events.json)	<code>log_tracking_event()</code>
55	Timestamped per-run campaign logs	<code>email_campaign_logs/</code>
56	Health-check <code>/status</code> on Flask & PHP	<code>status()</code>
57	Thread-lock file safety	<code>excel_lock</code>
58	Retry-on-file-lock & reconnect on SMTP drop	both scripts



Future releases from this foundation

The architecture was built to grow. Here is the prioritized release plan for the team's next cycles.

Product Roadmap

P1 → P3

P1 - High

W1 · Interactive Web DashboardFlask dashboard on top of Email_Tracking.xlsx — live charts for opens, clicks, bounces, reply rates; filters by date/country/language.

P1 - High

W2 · Unsubscribe & Suppression ManagerA central blocklist (opt-out) so unsubscribed leads are never re-contacted across campaigns.

P1 - High

W3 · Cloud Scheduler (cron/job)Move reply/reminder/sync loops to a scheduled job on cPanel or a VM so the laptop doesn't need to stay on.

P1 - High

W4 · Multi-campaign managerRun several lead files / segments with separate sheets and per-campaign Config, without editing code.

P2 - Medium

Multi-mailbox rotationRotate several sender addresses to spread volume and protect deliverability of the main mailbox.

P2 - Medium

Open-time + A/B subject splitSend two subject variants to a sample and pick the winner for the remainder.

P2 - Medium

Bounce-rate health alertsFlag when bounce rate crosses a threshold; notify via mail.

P2 - Medium

Per-lead timeline viewExpandable row in the dashboard showing send → open → click → visit → reply events in order.

P3 – Future

CRM & API integrations Sync replies/opens to HubSpot / Zoho / your CRM; REST API so other tools can push leads.

P3 – Future

Multi-channel follow-up Add WhatsApp / SMS / LinkedIn-touch based on the same tracked-event data.

P3 – Future

SaaS packaging Package the system as a configurable product with profiles, billing-ready dashboards and teams.

Workshop: surprising wins already inside

Already ready

- cloudflared + PHP tracker gives a **free public endpoint** for dev/testing
- Excel tracker doubles as a **hand-off report** to non-technical teammates
- All major logic is **already split into standalone scripts** (reply-checker, sync)

Needs a little work

- Password stored in plaintext → move to env vars / secure store
- IP geolocation is best-effort; could add a paid source for accuracy
- Batch-write Excel is fast but later scale warrants a proper DB

Technology & what the project ships

Built 100% on free, open source tools — this keeps the cost at zero for the company.

Technology Stack	
LAYER	TOOL
Language	Python 3 · PHP 7/8
Sending	<code>smtpplib</code> (SMTP :465 SSL)
Inbox	<code>imaplib</code> (IMAP :993 SSL)
Tracking HTTP	Flask :6060 + cloudflared
Prod tracker	PHP on cPanel (<code>track.php</code>)
Excel	<code>openpyxl</code> + <code>pandas</code>
Translation	<code>deep-translator</code> + <code>bs4</code>
Validation	<code>dnspython</code> + smtp probe
Geolocation	ip-api.com (free) + ipinfo.io
CLI / progress	<code>requests</code> · <code>tqdm</code>

Deliverables Shipped	
ARTIFACT	
1	Email Automation Pipeline — 4 Python scripts
2	PHP Tracking Backend — 3 deployment variants

- 3 Email Templates + Portfolio tracking snippet
- 4 Excel Master Tracker — 6 sheets, 25 columns
- 5 Auto-refreshed Sent/Bounced/Summary report
- 6 Full team documentation (12 markdown docs)
- 7 This report + flowcharts (HTML / PDF)

Success criteria (all met): opens recorded automatically · fake CEO emails caught before send · non-openers re-contacted once · replies scanned every 5 minutes · full audit trail across every run.

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